ATM TRANSACTION SYSTEM

Phase 2: SQL Queries on Designed Database

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Declaration

We hereby declare that this project titled **'ATM Transaction System – Phase 2: SQL Queries'** is an original work carried out by us under the guidance of Mr. Atul Kumar Singh. This project has not been submitted previously for any academic evaluation.

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1. Introduction

This report presents Phase 2 of the ATM Transaction System DBMS project. Building on the Entity-Relationship diagram and table creation from Phase 1, this phase focuses on populating the database with realistic sample data (10 records per table) and executing 15 meaningful SQL queries. The queries cover a range of operations including simple selection, pattern matching using LIKE and wildcards, aggregate functions, multi-table JOINs, WHERE filters, and ORDER BY / GROUP BY clauses.

2. Sample Data — INSERT Statements

The following INSERT statements populate each table with 10 records of realistic data.

2.1 BANK

```
INSERT INTO BANK VALUES
(1, 'SBIN0001234', 'State Bank of India', 'Mumbai'),
(2, 'HDFC0001234', 'HDFC Bank', 'Mumbai'),
(3, 'ICIC0001234', 'ICICI Bank', 'Mumbai'),
(4, 'PUNB0001234', 'Punjab National Bank', 'New Delhi'),
(5, 'UTIB0001234', 'Axis Bank', 'Mumbai'),
(6, 'BARB0001234', 'Bank of Baroda', 'Vadodara'),
(7, 'KKBK0001234', 'Kotak Mahindra Bank', 'Mumbai'),
(8, 'IOBA0001234', 'Indian Overseas Bank', 'Chennai'),
(9, 'CNRB0001234', 'Canara Bank', 'Bengaluru'),
(10,'UBIN0001234', 'Union Bank of India', 'Mumbai');
```

2.2 CUSTOMER

```
INSERT INTO CUSTOMER VALUES
(101, 'Aarav Sharma', '9811001001', '12 MG Road, Delhi', 'aarav@gmail.com'),
(102, 'Priya Mehta', '9822002002', '45 Park St, Mumbai', 'priya@yahoo.com'),
(103, 'Rahul Verma', '9833003003', '78 Anna Nagar, Chennai', 'rahul@outlook.com'),
(104, 'Sneha Gupta', '9844004004', '23 FC Road, Pune', 'sneha@gmail.com'),
(105, 'Karan Singh', '9855005005', '56 Brigade Rd, Bengaluru','karan@gmail.com'),
(106, 'Nisha Patel', '9866006006', '89 Lal Darwaza, Ahmedabad','nisha@gmail.com'),
(107, 'Vikram Joshi', '9877007007', '34 Civil Lines, Jaipur', 'vikram@gmail.com'),
(108, 'Anjali Rao', '9888008008', '67 Banjara Hills, Hyd', 'anjali@gmail.com'),
(109, 'Rohit Kumar', '9899009009', '11 Sector 17, Chandigarh','rohit@gmail.com'),
(110, 'Divya Nair', '9810010010', '90 Marine Drive, Kochi', 'divya@gmail.com');
```

2.3 BANK_ACCOUNT

```
INSERT INTO BANK_ACCOUNT VALUES (1001, 'Savings',
85000.00, 1); (1002, 'Current', 250000.00, 2);
(1003, 'Savings', 42000.50, 3); (1004, 'Savings',
60000.00, 4); (1005, 'Current', 175000.00, 5);
(1006, 'Savings', 30000.00, 6); (1007, 'Savings',
95000.75, 7); (1008, 'Current', 320000.00, 8);
(1009, 'Savings', 15000.00, 9); (1010, 'Savings',
72000.00, 10);
```

2.4 ATM_CARD

```
INSERT INTO ATM_CARD VALUES (4001, 1234, '2026-12-31', 'Debit',
'Active', 101, 1001); (4002, 5678, '2025-06-30', 'Debit',
'Expired', 102, 1002); (4003, 9012, '2027-03-31', 'Credit',
'Active', 103, 1003); (4004, 3456, '2026-09-30', 'Debit',
'Active', 104, 1004); (4005, 7890, '2024-12-31', 'Credit',
'Blocked', 105, 1005); (4006, 2345, '2027-06-30', 'Debit',
'Active', 106, 1006); (4007, 6789, '2026-03-31', 'Debit',
'Active', 107, 1007); (4008, 0123, '2025-09-30', 'Credit',
'Active', 108, 1008); (4009, 4567, '2027-12-31', 'Debit',
'Active', 109, 1009); (4010, 8901, '2026-06-30', 'Debit',
'Expired', 110, 1010);
```

2.5 ATM_MACHINE

```
INSERT INTO ATM_MACHINE VALUES (201, 'Connaught Place, Delhi', 'Online',
500000.00, 'Online', 'SBI', 1); (202, 'Andheri West, Mumbai', 'Online',
350000.00, 'Online', 'HDFC', 2); (203, 'T. Nagar, Chennai', 'Offline', 0.00,
'Offline', 'ICICI', 3); (204, 'Karol Bagh, Delhi', 'Online', 420000.00,
'Online', 'PNB', 4); (205, 'Koramangala, Bengaluru', 'Online', 280000.00,
'Online', 'Axis', 5); (206, 'Navrangpura, Ahmedabad', 'Online', 190000.00,
'Online', 'BOB', 6); (207, 'Vastrapur, Ahmedabad', 'Offline', 0.00,
'Offline', 'Kotak', 7); (208, 'Banjara Hills, Hyderabad', 'Online',
610000.00, 'Online', 'IOB', 8); (209, 'Sector 22, Chandigarh', 'Online',
230000.00, 'Online', 'Canara',9); (210, 'MG Road, Kochi', 'Online',
390000.00, 'Online', 'UBI', 10);
```

2.6 TRANSACTION

```
INSERT INTO TRANSACTION VALUES (301, 'Withdrawal', '2025-01-15 10:30:00',
5000.00, 'Success', 4001, 201); (302, 'Deposit', '2025-01-16 11:00:00',
10000.00, 'Success', 4002, 202); (303, 'Withdrawal', '2025-01-17 09:15:00',
2000.00, 'Failed', 4003, 203); (304, 'Balance', '2025-01-18 14:00:00', 0.00,
'Success', 4004, 204); (305, 'Withdrawal', '2025-01-19 16:45:00', 8000.00,
'Success', 4005, 205); (306, 'Withdrawal', '2025-01-20 08:30:00', 3000.00,
'Success', 4006, 206); (307, 'Deposit', '2025-01-21 12:00:00', 15000.00,
'Success', 4007, 207); (308, 'Withdrawal', '2025-01-22 17:30:00', 6000.00,
'Failed', 4008, 208); (309, 'Withdrawal', '2025-01-23 13:20:00', 4500.00,
'Success', 4009, 209); (310, 'Balance', '2025-01-24 10:00:00', 0.00, 'Success',
4010, 210);
```

3. Data Preview (Sample Records)

BANK

Bank_ID	IFSC_code	Bank_Name	Head_office
1	SBIN0001234	State Bank of India	Mumbai
2	HDFC0001234	HDFC Bank	Mumbai
3	ICIC0001234	ICICI Bank	Mumbai
4	PUNB0001234	Punjab National Bank	New Delhi
5	UTIB0001234	Axis Bank	Mumbai
6	BARB0001234	Bank of Baroda	Vadodara
7	KKBK0001234	Kotak Mahindra Bank	Mumbai
8	IOBA0001234	Indian Overseas Bank	Chennai
9	CNRB0001234	Canara Bank	Bengaluru
10	UBIN0001234	Union Bank of India	Mumbai

CUSTOMER

Customer_ID	Name	Phone	Address	Email
101	Aarav Sharma	9811001001	Delhi	aarav@gmail.com
102	Priya Mehta	9822002002	Mumbai	priya@yahoo.com
103	Rahul Verma	9833003003	Chennai	rahul@outlook.com
104	Sneha Gupta	9844004004	Pune	sneha@gmail.com
105	Karan Singh	9855005005	Bengaluru	karan@gmail.com
106	Nisha Patel	9866006006	Ahmedabad	nisha@gmail.com
107	Vikram Joshi	9877007007	Jaipur	vikram@gmail.com
108	Anjali Rao	9888008008	Hyderabad	anjali@gmail.com
109	Rohit Kumar	9899009009	Chandigarh	rohit@gmail.com
110	Divya Nair	9810010010	Kochi	divya@gmail.com

BANK_ACCOUNT

Acc_No	Acc_Type	Balance	Bank_ID
1001	Savings	85,000.00	1
1002	Current	2,50,000.00	2
1003	Savings	42,000.50	3
1004	Savings	60,000.00	4
1005	Current	1,75,000.00	5
1006	Savings	30,000.00	6

1007	Savings	95,000.75	7
1008	Current	3,20,000.00	8
1009	Savings	15,000.00	9
1010	Savings	72,000.00	10

ATM_CARD

Card_no	Expiry_Date	Card_Type	Card_Status	Customer_ID	Acc_No
4001	2026-12-31	Debit	Active	101	1001
4002	2025-06-30	Debit	Expired	102	1002
4003	2027-03-31	Credit	Active	103	1003
4004	2026-09-30	Debit	Active	104	1004
4005	2024-12-31	Credit	Blocked	105	1005
4006	2027-06-30	Debit	Active	106	1006
4007	2026-03-31	Debit	Active	107	1007
4008	2025-09-30	Credit	Active	108	1008
4009	2027-12-31	Debit	Active	109	1009
4010	2026-06-30	Debit	Expired	110	1010

ATM_MACHINE

ATM_ID	Location	STATUS	Cash_Avail	ATM_Status	Bank_ID
201	Connaught Place, Delhi	Online	5,00,000	Online	1
202	Andheri West, Mumbai	Online	3,50,000	Online	2
203	T. Nagar, Chennai	Offline	0	Offline	3
204	Karol Bagh, Delhi	Online	4,20,000	Online	4
205	Koramangala, Bengaluru	Online	2,80,000	Online	5
206	Navrangpura, Ahmedabad	Online	1,90,000	Online	6
207	Vastrapur, Ahmedabad	Offline	0	Offline	7
208	Banjara Hills, Hyderabad	Online	6,10,000	Online	8
209	Sector 22, Chandigarh	Online	2,30,000	Online	9
210	MG Road, Kochi	Online	3,90,000	Online	10

TRANSACTION

Trans_ID	Trans_type	Date_time	Amount	Status	Card_no	ATM_ID
301	Withdrawal	2025-01-15 10:30	5,000	Success	4001	201
302	Deposit	2025-01-16 11:00	10,000	Success	4002	202
303	Withdrawal	2025-01-17 09:15	2,000	Failed	4003	203
304	Balance	2025-01-18 14:00	0	Success	4004	204
305	Withdrawal	2025-01-19 16:45	8,000	Success	4005	205
306	Withdrawal	2025-01-20 08:30	3,000	Success	4006	206
307	Deposit	2025-01-21 12:00	15,000	Success	4007	207
308	Withdrawal	2025-01-22 17:30	6,000	Failed	4008	208
309	Withdrawal	2025-01-23 13:20	4,500	Success	4009	209
310	Balance	2025-01-24 10:00	0	Success	4010	210

4. SQL Queries with Expected Output

Query 1 Retrieve all records from the CUSTOMER table.

```
SELECT * FROM CUSTOMER;
```

Query 2 Fetch all ATM cards that currently have an 'Active' status.

```
SELECT Card_no, Card_Type, Expiry_Date, Card_Status
FROM ATM_CARD
WHERE Card_Status = 'Active';
```

Query 3 Find all customers whose email address ends with '@gmail.com'.

```
SELECT Name, Email
FROM CUSTOMER
WHERE Email LIKE '%@gmail.com';
```

Query 4 Find ATM cards with expiry year 2026 using wildcard pattern matching.

```
SELECT Card_no, Card_Type, Expiry_Date
FROM ATM_CARD
WHERE Expiry_Date LIKE '2026%';
```

Query 5 Retrieve savings accounts with a balance greater than 50,000.

```
SELECT Acc_No, Acc_Type, Balance
FROM BANK_ACCOUNT
WHERE Acc_Type = 'Savings' AND Balance > 50000;
```

Query 6 List all transactions ordered from the highest to the lowest amount.

```
SELECT Trans_ID, Trans_type, Amount, Status
FROM TRANSACTION
ORDER BY Amount DESC;
```

Query 7 Display customer name along with their linked ATM card number and card type.

```
SELECT C.Name, C.Phone_number, A.Card_no, A.Card_Type
FROM CUSTOMER C
INNER JOIN ATM_CARD A ON C.Customer_ID = A.Customer_ID;
```

Query 8 Show customer name, account balance, account type, and their bank name.

```
SELECT C.Name, BA.Acc_Type, BA.Balance, B.Bank_Name
FROM CUSTOMER C
JOIN ATM_CARD AC ON C.Customer_ID = AC.Customer_ID
JOIN BANK_ACCOUNT BA ON AC.Acc_No = BA.Acc_No
JOIN BANK B ON BA.Bank_ID = B.Bank_ID;
```

Query 9 List all successful withdrawal transactions with the ATM location.

```
SELECT T.Trans_ID, T.Amount, T.Date_time, M.Location
FROM TRANSACTION T
JOIN ATM_MACHINE M ON T.ATM_ID = M.ATM_ID
WHERE T.Trans_type = 'Withdrawal' AND T.Status = 'Success';
```

Query 10 Count how many transactions each ATM card has performed.

```
SELECT Card_no, COUNT(Trans_ID) AS Total_Transactions
FROM TRANSACTION GROUP BY Card_no ORDER BY
Total_Transactions DESC;
```

Query 11 Calculate the total transaction amount processed at each ATM machine.

```
SELECT ATM_ID, SUM(Amount) AS Total_Amount
FROM TRANSACTION
GROUP BY ATM_ID
ORDER BY Total_Amount DESC;
```

Query 12 Find ATM cards that have performed more than one transaction.

```
SELECT Card_no, COUNT(Trans_ID) AS Txn_Count
FROM TRANSACTION GROUP BY Card_no HAVING
COUNT(Trans_ID) > 1;
```

Query 13 Fetch all transactions done at ATM machines 201, 205, or 209.

```
SELECT Trans_ID, Trans_type, Amount, ATM_ID
FROM TRANSACTION
WHERE ATM_ID IN (201, 205, 209);
```

Query 14 Find all bank accounts whose balance falls between 50,000 and 2,00,000.

```
SELECT Acc_No, Acc_Type, Balance
FROM BANK_ACCOUNT
WHERE Balance BETWEEN 50000 AND 200000;
```

Query 15 Show the customer name, card number, and transaction details for all failed transactions.

```
SELECT C.Name, T.Card_no, T.Trans_type, T.Amount, T.Date_time
FROM TRANSACTION T
JOIN ATM_CARD AC ON T.Card_no = AC.Card_no
JOIN CUSTOMER C ON AC.Customer_ID = C.Customer_ID
WHERE T.Status = 'Failed';
```